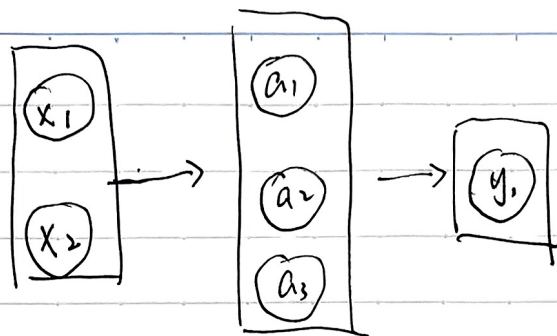


Q1

1.



$$\vec{a} = \sigma(\vec{w} \cdot \vec{x} + \vec{b})$$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \sigma \left(\begin{bmatrix} b_{x1} \\ b_{x2} \\ b_{x3} \end{bmatrix} + \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \\ w_{31} & w_{32} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right)$$

$$\vec{y} = \sigma(\vec{c}^T \vec{a})$$

$$y_1 = b + [c_1 \ c_2 \ c_3] \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$y = b + \sum_i c_i \sigma(b_i + \sum_j w_{ij} x_j)$$

2.

$$L(y, \hat{y}) = -[y \cdot \log(\hat{y}) + (1-y) \cdot \log(1-\hat{y})]$$

① 当真实标签 $y \geq 1$, $L(y, \hat{y}) = -\log(\hat{y})$, $\hat{y}$ 越接近 1, $L(y, \hat{y})$ 越小, $\hat{y}$ 越接近 0, $L(y, \hat{y})$ 越大;

② 当真实标签 $y = 0$, $L(y, \hat{y}) = \log(1-\hat{y})$, $\hat{y}$ 越接近 0, $L(y, \hat{y})$ 越小, $\hat{y}$ 越接近 1, $L(y, \hat{y})$ 越大。

Q2

1. 全初始化为 0:

$$\text{如 } Q_1(1), \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \\ w_{31} & w_{32} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, [c_1 \ c_2 \ c_3] = [0 \ 0 \ 0]$$

$$F = Z = WX, \quad z_1 = z_2 = z_3 = 0 \quad \therefore a_1 = a_2 = a_3 = \phi(0)$$

$$\therefore \text{梯度下降时, } \frac{\partial L}{\partial w} = \frac{\partial L}{\partial h} \cdot \frac{\partial h}{\partial w} = \frac{\partial L}{\partial h} \cdot \phi'(WX+b) \cdot x^T$$

$$\therefore W \cdot b \text{ 为 } 0 \quad \therefore \phi' \text{ 为固定值}$$

$$\text{当每个神经元梯度一样时, } \frac{\partial L}{\partial w_i} = \delta \cdot x = \frac{\partial L}{\partial h_i} \cdot \phi'(0) \text{ 均一样}$$

$\therefore$ 一层中权重一致, 使模型失效

全初始化为其他相同值: 假设为 m . $W = m \cdot 1, b = m \cdot 1$

$$WX+b = m \cdot (1 \cdot x) + m \cdot 1 = m \cdot 1. \text{ 为常数矩阵}$$

$$\therefore a = \sigma(WX+b) = \phi(m \cdot 1) = \phi(m) \cdot 1. \text{ 固定不变}$$

每个神经元输出相同, $\frac{\partial L}{\partial w} = \delta \cdot x^T$ 一致, 模型失效



2. $\frac{\partial L}{\partial w} = 2w + 2$, 当 $w_0 = 3$ 时, $\frac{\partial L}{\partial w} = 8$

$\therefore w_1 = w_0 - \eta \cdot \frac{\partial L}{\partial w} = 3 - 8\eta$, 向负梯度方向更新

Q3.

1. $1 - \sigma(x) = 1 - \frac{1}{1+e^x} = \frac{e^x}{1+e^x} = \frac{e^{-x} \cdot e^x}{e^x + e^{-x} \cdot e^x} = \frac{1}{e^x + 1}$

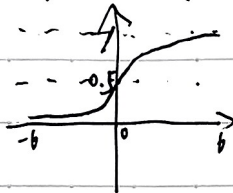
$\sigma(-x) = \frac{1}{1+e^x}$

$\therefore 1 - \sigma(x) = \sigma(-x)$

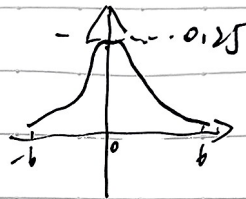
2. $\sigma'(x) = \frac{d}{dx} \left(\frac{1}{1+e^x} \right) = (-1) \cdot \frac{1}{(1+e^x)^2} \cdot e^x \cdot (1) = \frac{e^{-x}}{(1+e^x)^2}$

$\sigma(x)(1-\sigma(x)) = \frac{1}{1+e^x} \cdot \frac{e^{-x}}{1+e^x} = \frac{e^{-x}}{(1+e^x)^2}$

$\therefore \sigma'(x) = \sigma(x)(1-\sigma(x))$



$\sigma(x)$ 图像



$\sigma'(x)$ 函数

3. $\sigma(x)$ 函数值从 0 开始又很快趋向 1, 使得更新中权重变化很小, 使得收敛很慢; 在多层网络中, 梯度相乘导致梯度指数级衰减, 无法有效更新网络权重, 导致梯度消失。

